

Practice Set 1

1. Write a Program to ask Student for his Name & City Print both value with Proper formatting.

#Without Parameter

```
def student_details():  
    name = input("Enter Student Name: ")  
    city = input("Enter City: ")  
  
    print("Student Details")  
    print("Name :", name)  
    print("City :", city)
```

#With Parameter

```
def student_details(name, city):  
    print("Student Details")  
    print("Name :", name)  
    print("City :", city)
```

2. Write a program that takes two integers as input and prints their sum, difference, product, and quotient.

#Without Parameter

```
def arithmetic_operations():  
    a = int(input("Enter first number: "))  
    b = int(input("Enter second number: "))  
  
    print("Sum =", a + b)  
    print("Abstraction =", a - b)  
    print("Multiplication =", a * b)  
    print("Divison =", a / b)
```

```
#with Parameter
def arithmetic_operations(a, b):
    print("Sum =", a + b)
    print("Abstraction =", a - b)
    print("Multiplication =", a * b)
    print("Division =", a / b)
```

3. Create a program that calculates the area of a circle given its radius (use $\pi = 3.14$).

```
#Without Parameter
def area_of_circle():
    r = float(input("Enter radius of the circle: "))
    area = 3.14 * r * r
    print("Area of Circle =", area)

#With Parameter
def area_of_circle(r):
    area = 3.14 * r * r
    print("Area of Circle =", area)
```

4. Write a Program to check whether enter number is positive or negative

```
#without parameter
def check_number():
    num = int(input("Enter a number: "))

    if num > 0:
        print("Number is Positive")
    elif num < 0:
        print("Number is Negative")
    else:
        print("Number is Zero")
```

```
#with parameter
def check_number(num):
    if num > 0:
        print("Number is Positive")
    elif num < 0:
        print("Number is Negative")
    else:
        print("Number is Zero")
```

5. Write a program that swaps the values of two variables without using a third variable.

```
#without parameter
def swap_values():
    a = int(input("Enter first number: "))
    b = int(input("Enter second number: "))

    print("Before Swapping:")
    print("a =", a, "b =", b)
    print("After Swapping:")
    print("a =", b, "b =", a)

#with parameter
def swap_values(a, b):
    print("Before Swapping:")
    print("a =", a, "b =", b)
    print("After Swapping:")
    print("a =", b, "b =", a)
```

6. Write a program that prompts the user to enter their distance in Km and then converts it to Miles.

```
#without parameter
def km_to_miles():
    km = (input("Enter distance in kilometers: "))
    miles = km * 0.621371
    print("Distance in miles =", miles)
```

```
#with parameter
def km_to_miles(km):
    miles = km * 0.621371
    print("Distance in miles =", miles)
```

7. Write a program to Convert Celsius into Fahrenheit

```
#without parameter
def celsius_to_fahrenheit():
    c = float(input("Enter temperature in Celsius: "))
    f = (c * 9/5) + 32
    print("Temperature in Fahrenheit =", f)
```

```
#with parameter
def celsius_to_fahrenheit(c):
    f = (c * 9/5) + 32
    print("Temperature in Fahrenheit =", f)
```